

**VIET NAM UNIVERSITY
HANOI SCHOOL OF BUSINESS AND MANAGEMENT**



FINAL PROJECT REPORT

Course: HSB 2006 – Programming and IT Applications in Business

Project Title: PickleConnect – Pickleball Coaching Connection System

Class: MET 5

Members:

1. Nguyễn Thạch Bảo Long – 23080058 (Leader)
2. Trần Thu Lan – 23080051
3. Ngô Quý Khoa – 22080044
4. Nguyễn Minh Quân – 23080077
5. Nguyễn Thị Diễm Quỳnh – 23080078

Teacher: Dr. Ngô Trung Kiên

Hà Nội, 2026

TABLE OF CONTENT

LIST OF FIGURES	4
1. Introduction.....	5
2. Business Problem and Objectives.....	5
2.1 Business Problem.....	5
2.2 Project Objectives	5
2.3 Version Focus	5
3. Scope of the System.....	6
3.1 In Scope (Implemented in the current version).....	6
3.2 Out of Scope (Future development).....	6
4. Requirements.....	7
4.1 Functional Requirements	7
4.2 Non-Functional Requirements	8
4.3 Business Rules.....	8
5. User Stories.....	9
5.1 Student (PLAYER)	9
5.2 Coach (COACH)	9
5.3 Admin (ADMIN).....	10
5.4 Acceptance Criteria (key stories)	10
6. System Diagrams.....	11
6.1 Use Case Diagram.....	11
6.2 Activity Diagram	13
6.3 Sequence Diagram	14
6.4 Relationship among the three diagrams.....	15
7. System Architecture and Technology Stack.....	15
7.1 Architecture Overview.....	15
7.2 Technology Stack	16
7.3 Design Approach	16
8. Database Design	16
8.1 Database Overview	16
8.2 Main Tables	17
8.3 Key Relationships	17
8.4 Booking Data Rules	17
8.5 Design Notes.....	18

9. Implementation	18
9.1 Authentication and Authorization.....	18
9.2 Admin Module	18
9.3 Coach Module	19
9.4 Student Module and Booking Workflow	19
9.5 Search, Filter, and Sort	21
10. Security	21
11. Testing.....	22
11.1 Test Approach	22
11.2 Summary of Results	22
11.3 Main Defects Fixed	22
11.4 Conclusion of Testing.....	22
12. Installation Guide and Test Accounts.....	23
12.1 Requirements.....	23
12.2 Setup Steps.....	23
12.3 Test Accounts	23
12.4 Repository.....	23
13. Task Allocation.....	23
14. Conclusion and Future Development.....	24
14.1 Conclusion	24
14.2 Limitations	24
14.3 Future Development	24
15. References	25
16. AI / Tool-Use Declaration.....	25

LIST OF FIGURES

<i>Figure 1. Use Case Diagram.....</i>	<i>11</i>
<i>Figure 2. Activity Diagram</i>	<i>13</i>
<i>Figure 3. Sequence Diagram.....</i>	<i>14</i>
<i>Figure 4. Login/ logout hub.....</i>	<i>18</i>
<i>Figure 5. Admin interface.....</i>	<i>19</i>
<i>Figure 6. Coach module.....</i>	<i>19</i>
<i>Figure 7. User Interface.....</i>	<i>20</i>
<i>Figure 8. Coach Search Interface.....</i>	<i>20</i>
<i>Figure 9. Pickle ball Training Booking.....</i>	<i>20</i>
<i>Figure 10. My schedule.....</i>	<i>21</i>
<i>Figure 11. Booking sucessfully.....</i>	<i>22</i>

1. Introduction

PickleConnect is a web-based platform designed to connect pickleball players with verified coaches. The system supports three user roles — Student (Player), Coach, and Admin — and provides core functions such as authentication, coach profile management, training schedule management, and session booking.

The project is developed as a practical business web application using HTML/CSS/JavaScript for the frontend, PHP for the backend, and MySQL for the database. It runs on a local XAMPP environment and is structured for potential online deployment.

This report describes the business problem, system scope, requirements, design diagrams, implementation, testing, and deployment instructions of the current version.

2. Business Problem and Objectives

2.1 Business Problem

Many pickleball players currently search for coaches through social media groups. Information is often incomplete, difficult to verify, and not managed in a structured way. Players waste time checking coach experience, availability, and credibility. Coaches also lack a central place to present verified profiles and manage teaching schedules.

2.2 Project Objectives

The main objectives of this project are:

- Build a web system that connects students and pickleball coaches.
- Support role-based access for Student, Coach, and Admin.
- Allow coaches to manage profiles and teaching schedules.
- Allow students to search coaches and book training sessions.
- Allow admins to approve coaches and manage user accounts.
- Store all core data in a relational database with basic security controls.

2.3 Version Focus

The current version focuses on an MVP (Minimum Viable Product): authentication, authorization, coach approval, schedule management, and the complete training booking

workflow. Advanced features such as online payment, player matching by skill level, and learning resources are identified as future development.

3. Scope of the System

3.1 In Scope (Implemented in the current version)

The following features are implemented and can be demonstrated:

- User registration and login with three roles: ADMIN, COACH, and PLAYER
- Role-based access control and protected pages
- Coach profile management (basic profile and teaching schedule views)
- Admin dashboard with coach approval and user account overview
- Student home page with navigation to core functions
- Coach search with keyword search, price filter, and sorting
- Training session booking (view available slots, book a session)
- View personal bookings and cancel a confirmed booking
- MySQL database for users, coach profiles, courts, schedules, and bookings
- Basic security: password hashing, session management, and prepared statements
- System documentation: requirements, user stories, diagrams, test plan, and test report

3.2 Out of Scope (Future development)

The following items are described in the original vision but are not fully implemented in this version:

- Online payment and escrow handling
- Featured coach packages with paid promotion
- Detailed star-rating review UI after completed sessions
- Learning resources / technical training materials
- Player matching by skill level (matchmaking)
- Mobile application
- Advanced analytics and revenue reporting

These items are reserved for future development and are not claimed as completed features in the current submission.

4. Requirements

4.1 Functional Requirements

FR01. The system shall allow users to register and log in with one of three roles: Student (PLAYER), Coach (COACH), or Admin (ADMIN).

FR02. The system shall redirect each user to the correct home page based on role after successful login.

FR03. The system shall prevent unauthorized access to protected pages when the user is not logged in.

FR04. The system shall prevent a user of one role from accessing pages belonging to another role.

FR05. Students shall be able to search coaches by keyword (name or profile description).

FR06. Students shall be able to filter coaches by hourly rate range.

FR07. Students shall be able to sort coaches by rating, price, or name.

FR08. Students shall be able to view available training time slots.

FR09. Students shall be able to book an available training session.

FR10. Students shall be able to view their own booking history.

FR11. Students shall be able to cancel a confirmed booking; the related time slot shall become available again.

FR12. The system shall not allow booking of a time slot that is already booked.

FR13. Coaches shall be able to view and manage their profile information (experience, rate, bio, certificates).

FR14. Coaches shall be able to view and manage teaching schedule entries.

FR15. Admins shall be able to view a basic dashboard with summary statistics.

FR16. Admins shall be able to review and approve or reject coach profiles.

FR17. Admins shall be able to view and manage user accounts (including lock/unlock actions at UI level).

FR18. The system shall store core data in a relational database (users, profiles, courts, schedules, bookings).

FR19. The system shall support logout and clear the user session.

FR20. The system shall display appropriate success and error messages for login and booking actions.

4.2 Non-Functional Requirements

NFR01. The system shall be a web application accessible through a modern browser (e.g., Chrome).

NFR02. The system shall run on a local server environment (XAMPP: Apache + MySQL + PHP).

NFR03. User passwords shall be stored using secure hashing (password_hash / password_verify).

NFR04. Database queries that use user input shall use prepared statements to reduce SQL injection risk.

NFR05. Authenticated pages shall rely on server-side session checks.

NFR06. The user interface shall be clear and usable for basic business operations.

NFR07. Core operations (login, booking, viewing lists) shall respond within an acceptable time on local deployment.

NFR08. The project source code and documentation shall be managed on GitHub with a clear folder structure.

NFR09. The system shall be documented with requirements, diagrams, test plan, and installation instructions.

NFR10. The interface shall use consistent styling across main modules (shared CSS + Bootstrap).

4.3 Business Rules

BR01. Only coaches marked as verified/approved should be treated as trusted coaches in the student-facing list where verification is applied.

BR02. A schedule slot with status AVAILABLE can be booked; a BOOKED slot cannot be booked again.

BR03. Each account has exactly one role at a time (ADMIN, COACH, or PLAYER).

BR04. A student can cancel only their own confirmed bookings.

BR05. When a booking is cancelled, the related schedule slot returns to AVAILABLE.

5. User Stories

5.1 Student (PLAYER)

US01. As a student, I want to register and log in so that I can access the system securely.

US02. As a student, I want to open my home page after login so that I can quickly reach main functions.

US03. As a student, I want to search coaches by name or description so that I can find a suitable coach.

US04. As a student, I want to filter coaches by price range so that I can stay within my budget.

US05. As a student, I want to sort coaches by rating, price, or name so that I can compare options easily.

US06. As a student, I want to view available training slots so that I can choose a suitable time.

US07. As a student, I want to book an available training session so that I can schedule practice with a coach.

US08. As a student, I want to view my booking history so that I can track my reserved sessions.

US09. As a student, I want to cancel a confirmed booking so that I can free a slot when my plan changes.

US10. As a student, I want to log out so that my account session is closed safely.

5.2 Coach (COACH)

US11. As a coach, I want to log in and open the coach page so that I can manage my work area.

US12. As a coach, I want to update my profile (experience, hourly rate, certificates, bio) so that students can understand my background.

US13. As a coach, I want to create and view teaching schedule slots so that students can book available times.

US14. As a coach, I want to see schedule status (available/booked) so that I can manage my teaching time.

5.3 Admin (ADMIN)

US15. As an admin, I want to log in to the admin dashboard so that I can supervise the platform.

US16. As an admin, I want to view basic statistics (coaches, students, pending coaches) so that I can monitor system status.

US17. As an admin, I want to approve or reject coach profiles so that only trusted coaches are listed.

US18. As an admin, I want to view user accounts by role so that I can manage platform users.

US19. As an admin, I want to lock or manage user access at account level so that I can control system safety.

5.4 Acceptance Criteria (key stories)

US07 – Book a training session

- Student must be logged in with role PLAYER
- Only slots with status AVAILABLE are shown
- After booking, a record is created in bookings
- The selected schedule status changes to BOOKED
- The same slot cannot be booked again

US09 – Cancel a booking

- Student can cancel only their own CONFIRMED booking
- Booking status changes to CANCELLED
- Related schedule status returns to AVAILABLE

US17 – Approve coaches

- Admin can see pending coach profiles
- Admin can approve or reject
- Approval status is reflected in the admin interface

US03–US05 – Search / Filter / Sort

- Student can search by keyword
- Student can filter by min/max price
- Student can sort by rating, price, or name

- Results update according to selected criteria

6. System Diagrams

This section presents three core diagrams used to analyze and design the PickleConnect system: Use Case Diagram, Activity Diagram, and Sequence Diagram.

6.1 Use Case Diagram

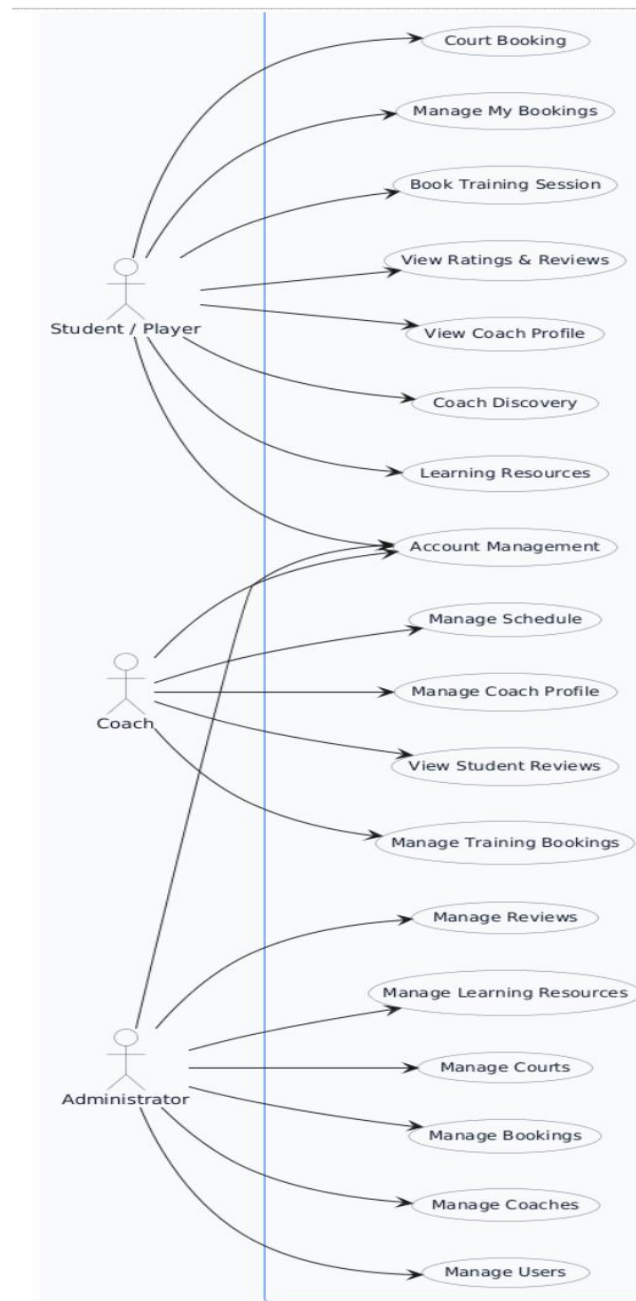


Figure 1. Use Case Diagram

The Use Case Diagram shows the main interactions between three actors and the system:

- Student / Player: search coaches, view profiles, book training sessions, manage personal bookings, and manage account access
- Coach: manage coach profile, manage teaching schedule, and view related booking information
- Administrator: manage users, manage coaches (approval), and supervise bookings / system data

The diagram provides a functional overview of what each role can do. Some extended use cases (for example court booking as a separate product line, or learning resources) are kept in the design vision; the implemented MVP prioritizes authentication, coach management, and training session booking.

6.2 Activity Diagram

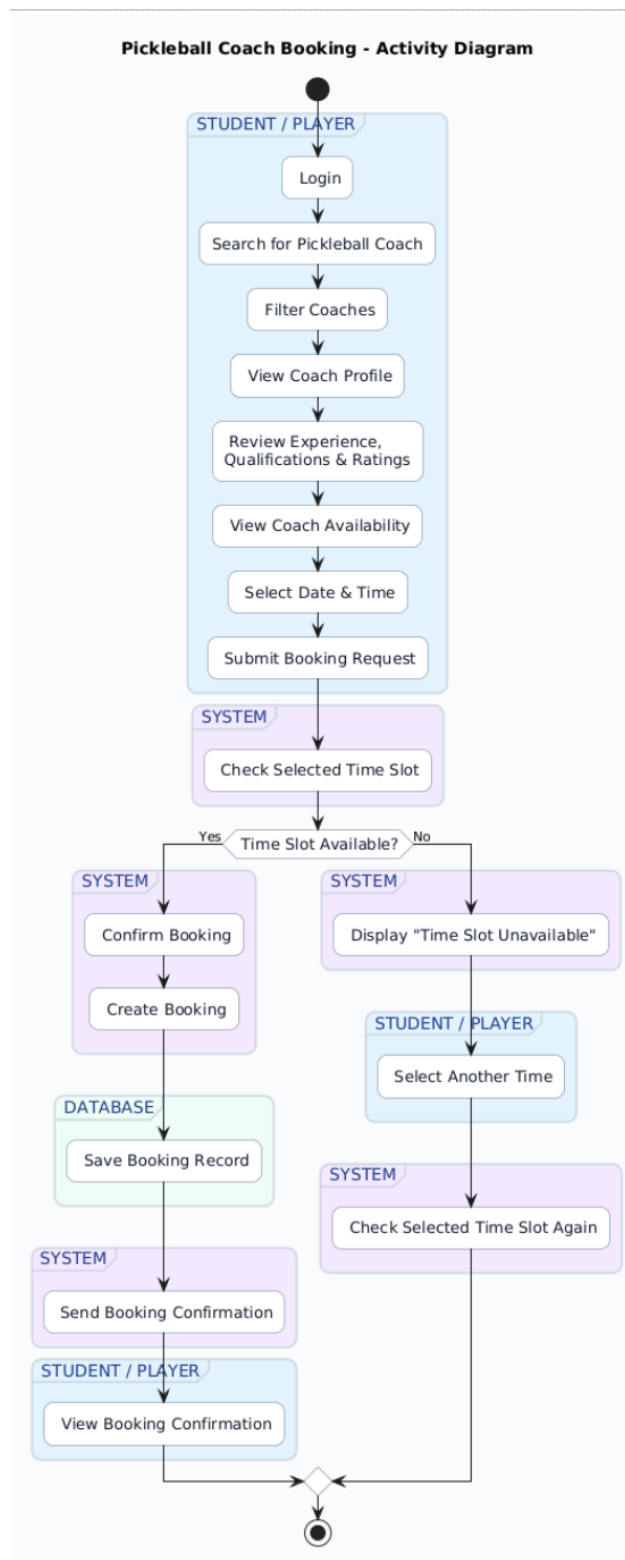


Figure 2. Activity Diagram

The Activity Diagram describes the end-to-end process of booking a training session:

- Student logs in
- Student searches and filters coaches

- Student views coach profile and availability
- Student selects date/time and submits a booking request
- System checks whether the selected time slot is available
- If available, the system confirms the booking and saves the record
- If unavailable, the system notifies the student to choose another slot

This diagram represents the main business workflow of the current system version.

6.3 Sequence Diagram

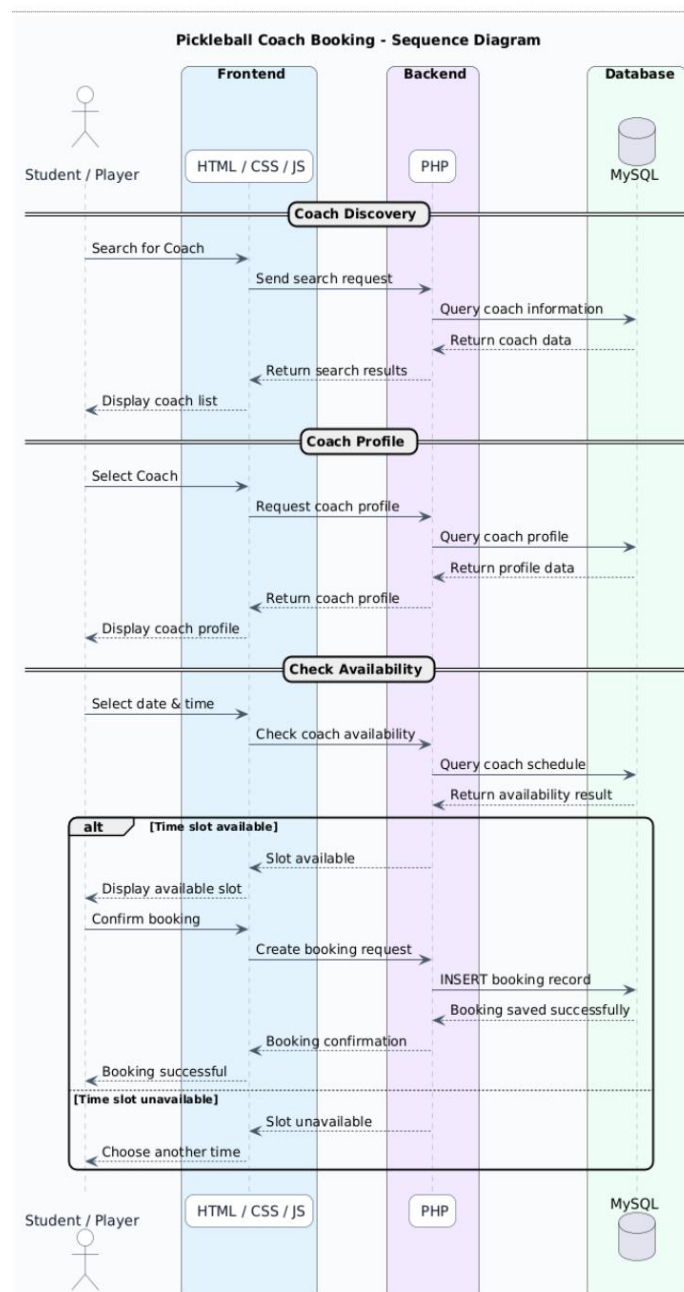


Figure 3. Sequence Diagram

The Sequence Diagram shows technical interactions among:

- Student (actor)
- Frontend (HTML/CSS/JS)
- Backend (PHP)
- Database (MySQL)

Main interaction groups:

- Coach discovery: search request → query coach data → return results
- Coach profile: request profile → query profile → display profile
- Availability and booking: check schedule → if available, insert booking and update slot status; if unavailable, return an error message

The diagram clarifies how the booking process is handled across presentation, application, and data layers.

6.4 Relationship among the three diagrams

- Use Case Diagram answers: What can each role do?
- Activity Diagram answers: What is the step-by-step business process of booking?
- Sequence Diagram answers: How do system components communicate to complete booking?

Together, the three diagrams connect requirements analysis, business process design, and technical implementation for the core booking feature.

7. System Architecture and Technology Stack

7.1 Architecture Overview

PickleConnect follows a classic three-layer web architecture:

- Presentation layer: HTML, CSS, JavaScript (Bootstrap-based UI)
- Application layer: PHP scripts handling authentication, authorization, booking, and admin/coach functions
- Data layer: MySQL database accessed through mysqli with prepared statements

The application is deployed locally using XAMPP (Apache + MySQL + PHP) and organized by role-based modules:

- /login.php, /register.php, /logout.php
- /admin/
- /coach/
- /student/
- /database/
- /doc/
- /testing/

7.2 Technology Stack

Layer	Technology
-----	-----
Frontend	HTML5, CSS3, JavaScript, Bootstrap 5
Backend	PHP 8.x
Database	MySQL
Local server	XAMPP (Apache, MySQL)
Version control	GitHub
Security support	PHP sessions, password hashing, prepared statements

7.3 Design Approach

The system is implemented as an MVP focused on a complete business workflow (training booking) with clear role separation. UI modules for Admin and Coach were integrated from team frontend work and connected to the shared authentication and database layer.

8. Database Design

8.1 Database Overview

The system uses a MySQL database named pickleball_db. The schema supports user accounts, role-based profiles, courts, teaching schedules, and bookings.

8.2 Main Tables

| Table | Purpose |

|-----|-----|

| users | Stores account information: email, password hash, full name, role (ADMIN/COACH/PLAYER), active status |

| coach_profiles | Stores coach professional data: experience, hourly rate, certificates, bio, rating, verification status |

| player_profiles | Stores additional student profile data (if used) |

| courts | Stores court/location information used in teaching schedules |

| schedules | Stores coach teaching slots: coach, court, start time, end time, price, status (AVAILABLE/BOOKED) |

| bookings | Stores student reservations linked to a schedule slot and booking/payment status |

| reviews | Designed for student ratings/reviews after sessions (schema ready; full UI is limited in MVP) |

8.3 Key Relationships

- One user has one role and may have one coach profile or player profile
- One coach (user) has many schedule slots
- One court can be used in many schedule slots
- One schedule slot can have one successful active booking path in the current workflow
- One student (player) can have many bookings
- Reviews are linked to coaches/students for future rating features

8.4 Booking Data Rules

- A schedule with status AVAILABLE can be booked
- After a successful booking, schedule status becomes BOOKED
- If a student cancels a confirmed booking, schedule status returns to AVAILABLE

- Booking records keep history even after cancellation (status updated to CANCELLED)

8.5 Design Notes

The schema is normalized for core operations: authentication, coach listing, schedule management, and booking. Some extended business ideas (escrow, featured packages, matchmaking) are not represented as full production tables in this version and remain for future work.

9. Implementation

9.1 Authentication and Authorization

The system provides login, logout, and role-based redirection.

- Login validates email/password against the users table
- Passwords are verified with password_verify against hashed values
- Successful login stores user_id, role, and full_name in PHP session
- Each protected page checks session and role before rendering content
- Unauthorized access is redirected to login.php

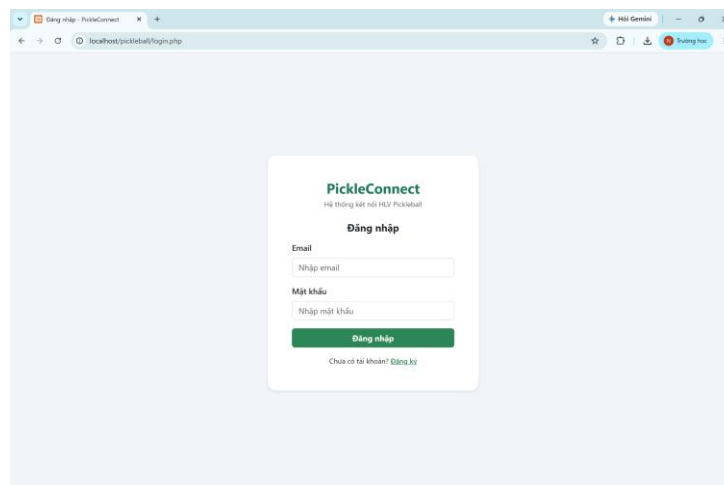


Figure 4. Login/ logout hub

9.2 Admin Module

The admin module supports platform supervision:

- Dashboard overview (coaches, students, pending coaches)

- Coach approval / management views
- User account listing by role

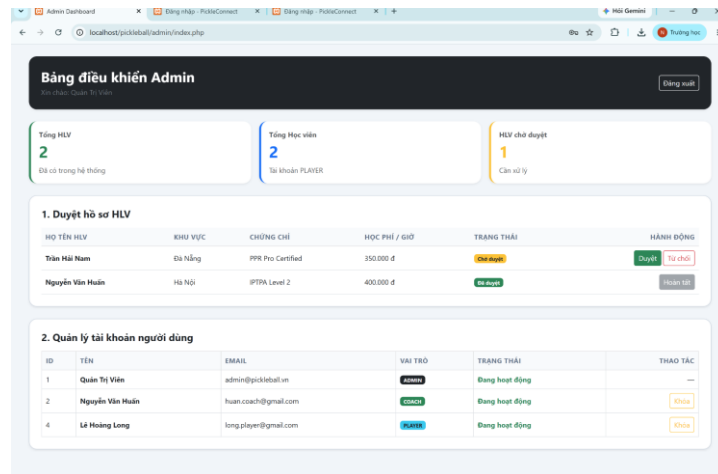


Figure 5. Admin interface

9.3 Coach Module

The coach module supports professional profile and schedule management:

- View/update coach profile information
- Manage teaching schedule entries and status

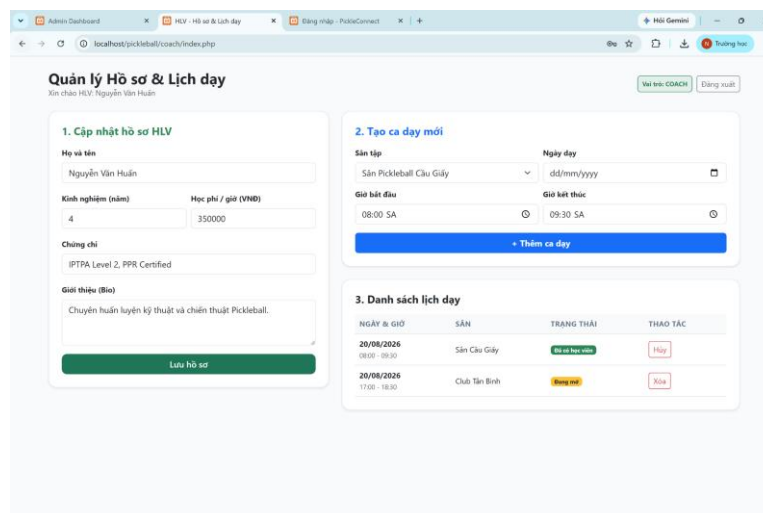


Figure 6. Coach module

9.4 Student Module and Booking Workflow

The student module implements the core business process:

- Student home with links to Find Coach, Book, and My Bookings
- Find Coach: keyword search, price filter, and sorting

- Book page: list AVAILABLE slots and submit booking
- My Bookings: view history and cancel confirmed bookings

Booking logic:

- Check selected schedule is still AVAILABLE
- Insert booking record
- Update schedule status to BOOKED inside a transaction
- Prevent double booking of the same slot
- Allow cancellation and restore slot availability

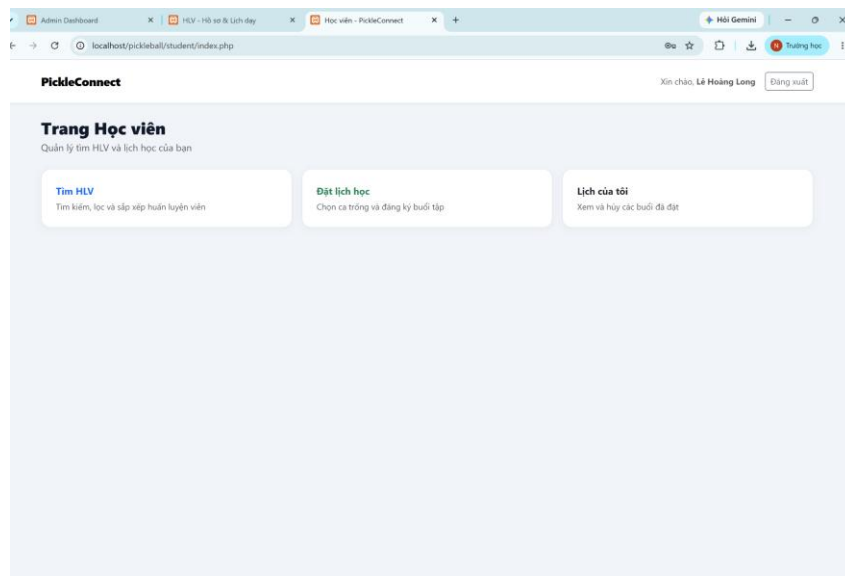


Figure 7. User Interface

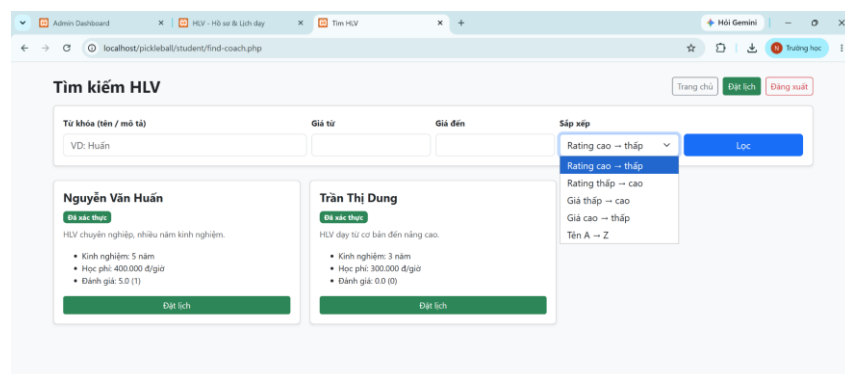


Figure 8. Coach Search Interface

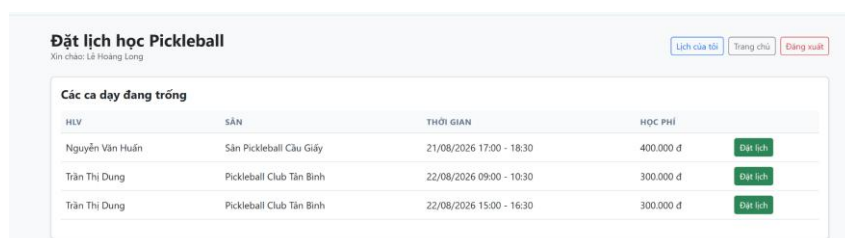


Figure 9. Pickle ball Training Booking

HLV	Sân	Thời gian	Học phí	Trạng thái	Thanh toán
Trần Thị Dung	Pickleball Club Tân Bình	22/08/2026 09:00 - 10:30	300.000 đ	CONFIRMED	UNPAID Hủy
Nguyễn Văn Huân	Sân Pickleball Cầu Giấy	21/08/2026 08:00 - 09:30	400.000 đ	CONFIRMED	UNPAID Hủy

Figure 10. My schedule

9.5 Search, Filter, and Sort

Students can:

- Search coaches by name/bio keyword
- Filter by minimum and maximum hourly rate
- Sort by rating, price, or name

This satisfies the required search/filter/sort capability of the course criteria.

10. Security

The current version applies the following controls

- Password hashing with PHP password_hash / password_verify
- Server-side session authentication for protected pages
- Role checks before accessing Admin, Coach, or Student pages
- Prepared statements for input-bound SQL queries
- Basic form confirmation for critical actions (book/cancel)
- Logout clears session data

Input handling and authorization checks are implemented on the server side. Advanced payment security is out of scope because online payment is not included in this version.

11. Testing

11.1 Test Approach

Testing followed the process: Test Plan → Execution → Defect → Fix → Retest → Report.

A full web-flow test set was prepared for authentication, authorization, booking, cancellation, and role pages.

11.2 Summary of Results

Item	Result
-----	-----
Total test cases	20
Failed in first run	2 (double-booking related)
Fixed and retested	2
Final result	20/20 Passed

11.3 Main Defects Fixed

1. Re-booking an already BOOKED slot
 - Fix: enforce AVAILABLE check and conditional UPDATE
2. Concurrent double-booking risk
 - Fix: wrap check + insert + update in a database transaction

11.4 Conclusion of Testing

Core flows (login by role, protected pages, book, cancel, search/filter/sort) are stable for demo and submission.



Figure 11. Booking sucessfully

12. Installation Guide and Test Accounts

12.1 Requirements

- Windows PC
- XAMPP (Apache + MySQL + PHP)
- Modern browser (Chrome recommended)

12.2 Setup Steps

1. Install and start XAMPP (Apache + MySQL)
2. Copy project source into:
C:\xampp\htdocs\pickleball\
3. Open phpMyAdmin → create database pickleball_db
4. Import database/pickleball.sql
5. Open:
<http://localhost/pickleball/login.php>

12.3 Test Accounts

Role	Email	Password
Admin	admin@pickleball.vn	password
Coach	huan.coach@gmail.com	password
Student	long.player@gmail.com	password

12.4 Repository

GitHub repository:

<https://github.com/Baolong23080058/B-i-cu-i-k-Nh-m-5->

13. Task Allocation

Member	Main responsibilities	Status
-----	-----	-----

| Nguyễn Thạch Bảo Long | Leader; Auth; booking backend; integration; GitHub structure; final report compilation | Completed |

| Trần Thu Lan | Student frontend design and UI contribution | Completed |

| Ngô Quý Khoa | Admin and Coach UI contribution | Completed |

| Nguyễn Minh Quân | Database design support; testing plan/report process | Completed |

| Nguyễn Thị Diễm Quỳnh | Scope, requirements, user stories, diagrams, report analysis sections, README support | Completed |

14. Conclusion and Future Development

14.1 Conclusion

PickleConnect delivers a working MVP for connecting pickleball students and coaches. The system implements authentication, role-based authorization, coach management, schedule handling, and a complete booking workflow with search/filter/sort support. Documentation includes requirements, diagrams, testing, and installation instructions. The implemented features are demonstrable on a local XAMPP environment.

14.2 Limitations

- Online payment and escrow are not implemented
- Full review UI and learning-resource modules are limited/not completed
- Player matchmaking by skill level is not implemented
- UI consistency is good for MVP but can be further refined

14.3 Future Development

- Online payment integration
- Post-session rating and review UI
- Featured coach packages
- Learning materials module
- Skill-based player matching
- Deployment to online hosting

15. References

- Course specification: HSB2006 – Developing Business Applications
- Bootstrap 5 documentation: <https://getbootstrap.com/>
- PHP Manual: <https://www.php.net/manual/en/>
- MySQL Documentation: <https://dev.mysql.com/doc/>
- XAMPP: <https://www.apachefriends.org/>
- Project repository: <https://github.com/Baolong23080058/B-i-cu-i-k-Nh-m-5->

16. AI / Tool-Use Declaration

AI/tool name	Purpose of use	Main parts assisted	How the output was verified
Generative AI assistants (e.g., ChatGPT / Grok)	Drafting report text, suggesting code structure, debugging guidance	Report wording, booking PHP snippets, test documentation structure	All features were implemented/tested on local XAMPP; code reviewed and adjusted by the team; only demonstrable functions are claimed

The team remains responsible for the correctness, security, and originality of the submitted work. All members can explain the implemented modules during demo/viva.